PAIGE MARUSCHAK

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FrontierTech Atlas

PHYSICS | FINANCE | EMERGING TECHNOLOGY

PROFILE

Student researcher and independent investor working across physics, computational materials, finance, and emerging technology. Experience includes single-crystal growth, low-temperature transport measurements, molecular-dynamics simulations, Python and Linux, investment research, technical presentations, and website development. Interested in understanding how advanced technologies work and evaluating how they can become viable businesses.

EDUCATION

2027 Graduate | Rolla High School

- Graduating one year early with a 3.91 GPA.

Dual-Enrolled Student | Missouri University of Science and Technology | **2025–Present**

- Completed Calculus I, II, and III with a 3.5 university GPA; Differential Equations and Linear Algebra planned for Spring 2027.

RESEARCH EXPERIENCE

Undergraduate Researcher | Missouri University of Science and Technology | **October 2025–Present**

NPT Barostat limitations in Molecular-Dynamics Simulations

- Tested the limits of the NPT barostat to get familiar with LAMMPS and the Mill, MS&T's local high performance computer.
- Used Linux, and C++

Undergraduate Researcher | Missouri University of Science and Technology | **Summer 2025**

Single-Crystal Growth and Physical Properties of Tb6Cr4Al43 | Presented July 27, 2025

- Completed approximately 250 hours of experimental research and produced a usable Tb6Cr4Al43 single crystal for structural and magnetic characterization.
- Investigated how the compound's atomic arrangement and rare-earth/transition-metal interactions influence magnetic ordering, anisotropy, and transport behavior.
- Used X-ray diffraction to study the unit cell and compare it with the Ho6Mo4Al43 structure.
- Prepared resistivity samples and studied transport properties in a superconducting magnet system.
- Temperature- and field-dependent resistivity measurements were interpreted as showing ferromagnetic behavior.
- Observed a residual resistivity ratio near 1, indicating little change in resistivity between room and low temperature.
- Measured increasing resistivity with magnetic field, consistent with normal-metal behavior; Hall-resistivity measurements were inconclusive because strong magnetic attraction interfered with the measurement.
- Delivered a 30-minute technical presentation describing the project, methods, and findings to graduate students.

Computational Materials Research | Li3PS4 Vacancy Configurations | **In Progress** | *Results presentation planned for May 2027 at Undergraduate research conference*

- Predicting the locations of the Lithium ions vacancies by creating a small neural network based on the configuration of PS4
- Optimizing the PS4 to make the Li ions transfer easier. This will help to make charging batteries quicker, and their lifetime will last longer.

FINANCE AND INVESTMENT

Independent Investor and Market Researcher | Self-Funded Fidelity Account

- Opened a brokerage account at age 15 using money earned by selling personal belongings; selected VTI as the first investment to prioritize diversification while developing investment knowledge.
- Manage a personal portfolio using company and sector research, market conditions, valuation, catalysts, and downside-risk analysis.
- Used the BUR investment and market response to the YPF litigation to study event risk, concentration, and the difference between apparent undervaluation and adequately priced uncertainty.

Financial Research Mentorship | Raymond James | **Monthly**

- Meet monthly with Ms. Sheila King, a lead portfolio analyst, to discuss emerging markets and technologies, potential investments, company health, private markets, corporate finance, private debt, inflation, labor markets, and macroeconomics.

Quantitative Finance Class| Stevens Institute of Technology | **July 2026**

- Took a quantitative finance class through Steven's Institute of Technology and received Bloomberg Market Concepts certification and a Capital IQ certification.

OpenAI and Public Markets Study

- Investigated how OpenAI-related announcements and commercial relationships corresponded with public-company returns, abnormal trading volume, and market-index contribution.

Conclusion was inconclusive to slightly positive.

- Examined event timing and company relationships while recognizing that limited contract disclosures constrained causal conclusions.

Constellation Energy Investment Pitch | Perplexity Stock Pitch Competition

- Connected nuclear-generation assets, AI-driven electricity demand, valuation, catalysts, market expectations, and risks in a technical-commercial investment thesis.

TECHNOLOGY AND DESIGN PROJECTS

FrontierTech Atlas | Next.js Website

- Designed and built a field guide to emerging energy systems, quantum-computing platforms, fuels, and materials, with detailed explanations of how each technology works and why it matters.
- Tracks important technical and commercial developments and offers multiple visual experiences, including Galactic, Art Deco, and Frontier views.
  - **Website:** alt-energy-database.vercel.app

Personal Résumé Website | https://paigemaruschak.vercel.app/

- Building an interactive portfolio that follows a particles-to-markets story, from particle physics and fusion energy to quantum computing and technology commercialization.
- Will allow visitors to view the résumé through Physics, Finance, and combined Physics × Finance perspectives.

Binary Clock | Engineering Project | In Development

- Designing a 5-by-15 binary clock with hour, minute, and second displays to build experience with electronics, digital logic, and physical computing.

EMPLOYMENT

Sandwich Artist | Jersey Mike's | **June 2026–Present**

- Work full time during summer and part time during the school year; serve customers, coordinate with coworkers, fill orders accurately and efficiently, creates cold and hot subs.

LEADERSHIP AND ACTIVITIES

Science Olympiad | Rolla High School | **4 Years**

- Earned second place in astronomy and third place in meteorology; also competed in wind power and got first place, experimental design in fourth place, and bridge design. Competed at reginal and state levels, won at regionals.

Speech and Debate | Rolla High School | **1 Year**

- International Extemporaneous Speaking: researched under time pressure and presented potential solutions.
- Informative Speaking: presentation about the W boson and the Standard Model using LEGO pieces.

Spanish Club | Rolla High School | **1 Year**

- Participated in club activities, celebrations, and cultural-learning events.

Society of Physics Students | Missouri S&T | **3 Years**

- Participate in physics discussions with guest lecturers and department professors and support activities including bake sales and fun Fermi questions.

Biomedical Engineering Prosthetics Design Team | Missouri S&T

- Serve as lead programmer and support CAD development for a myoelectric prosthetic arm intended to reduce weight, increase functionality, and improve affordability.

COMMUNITY SERVICE

Childcare Volunteer | Weekly | **6 Years**

- Work with children with special needs and teach lessons to students from kindergarten through fifth grade.

Rolla Celebration of Nations Volunteer | 1 Year

- Represented England at a multicultural festival hosted by Missouri S&T and the City of Rolla, sharing cultural information with attendees and helping with event passports.

CURRENT INTERESTS

Emerging technologies, AI and AI infrastructure, fusion, quantum computing, economics & finance.